260713

natu33

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Part I. Single-sequence OD model (non-block)

Indexing. A path is $x_0 = (x_0^1, \dots, x_0^{|x|})$ with origin $O = x_0^1$ and destination $D = x_0^{|x|}$; i indexes a *position* in the path, k indexes a vertex/state, and x_t^i is position i at diffusion step t.

Heat kernel.

$$C_t = e^{(A-D)\beta_t}, \qquad \bar{C}_t = e^{(A-D)\sum_{s \le t} \beta_s}. \tag{1}$$

Conditional denoiser. Given the noised path x_t, step t, and OD pair (O, D),

$$\hat{x}_0 = \mathrm{nn}_\theta\big(x_t, t, O, D\big). \tag{2}$$

For brevity write the model condition as $c \equiv \big(x_t, t, O, D\big)$.

Per-position posteriors.

$$\text{(true)} \quad \mathbf{p}^{\mathrm{post},i} \propto \big(x_t^i C_t\big) \odot \big(x_0^i \bar{C}_{t-1}\big), \tag{3}$$

$$\text{(model)} \quad \hat{\mathbf{p}}^{\,i} \propto \big(x_t^i C_t\big) \odot \big(\hat{x}_0^i(c)\, \bar{C}_{t-1}\big). \tag{4}$$

The true posterior does not depend on (O, D); the model posterior does, through $\hat{x}_0$.

Reverse-process KL (trains the denoiser).

$$\mathcal{L}_{\mathrm{KL}}^{(t)} = \sum_{\substack{i \\ i \notin \mathcal{C}}} \sum_{k=1}^{|V|} \big[\mathbf{p}^{\mathrm{post},i}\big]_k \log \frac{\big[\mathbf{p}^{\mathrm{post},i}\big]_k}{\big[\hat{\mathbf{p}}^{\,i}\big]_k}. \tag{5}$$

Clean-path cross-entropy (auxiliary).

$$\mathcal{L}_{\mathrm{CE}}^{(t)} = \sum_{\substack{i \\ i \notin \mathcal{C}}} \mathrm{CE}\big(x_0^i,\ p_\theta(\hat{x}_0^i \mid c)\big). \tag{6}$$

Connectivity penalty (graph validity).

$$\mathcal{L}_{\mathrm{CN}}^{(t)} = -\sum_{i=1}^{|x|-1} \sum_{(\hat{v}_0^i,\, \hat{v}_0^{i+1}) \in E} p_\theta(\hat{v}_0^i \mid c) \, \log p_\theta(\hat{v}_0^{i+1} \mid c). \tag{7}$$

Total objective.

$$\mathcal{L}(\theta) = \mathbb{E}_{(x_0, O, D) \sim \mathcal{P}} \, \mathbb{E}_{t \sim \mathcal{U}[1,T]} \, \mathbb{E}_{x_t \sim q(\cdot \mid x_0)} \Big[\mathcal{L}_{\mathrm{KL}}^{(t)} + \lambda \, \mathcal{L}_{\mathrm{CE}}^{(t)} + \mu \, \mathcal{L}_{\mathrm{CN}}^{(t)} \Big]. \tag{8}$$

Part II. Block-diffusion extension

Indexing. The path is split into blocks $x_0 = (\mathbf{x}_0^{(1)}, \ldots, \mathbf{x}_0^{(B)})$; b indexes blocks, i indexes a *position within block* b, and $\mathbf{x}_t^{(b),i}$ denotes position i of the noised block $\mathbf{x}_t^{(b)}$ at step t. The prefix $\mathbf{x}_0^{(<b)}$ collects all earlier (clean) blocks. The kernel $C_t, \bar{C}_t$ is unchanged from Part I.

Conditional denoiser. Given the noised block $\mathbf{x}_t^{(b)}$, step t, clean prefix $\mathbf{x}_0^{(<b)}$, and OD pair (O, D),

$$\hat{\mathbf{x}}_0^{(b)} = \mathrm{nn}_\theta\big(\mathbf{x}_t^{(b)}, t, \mathbf{x}_0^{(<b)}, O, D\big), \qquad c_b \equiv \big(\mathbf{x}_t^{(b)}, t, \mathbf{x}_0^{(<b)}, O, D\big). \tag{9}$$

Per-position posteriors.

$$\text{(true)} \quad \mathbf{p}^{\mathrm{post},(b,i)} \propto \big(\mathbf{x}_t^{(b),i} C_t\big) \odot \big(\mathbf{x}_0^{(b),i} \bar{C}_{t-1}\big), \tag{10}$$

$$\text{(model)} \quad \hat{\mathbf{p}}^{\,(b,i)} \propto \big(\mathbf{x}_t^{(b),i} C_t\big) \odot \big(\hat{\mathbf{x}}_0^{(b),i}(c_b) \, \bar{C}_{t-1}\big). \tag{11}$$

The true posterior does not depend on (O, D); the model posterior does, through $\hat{\mathbf{x}}_0$.

Reverse-process KL (trains the denoiser).

$$\mathcal{L}_{\mathrm{KL}}^{(b,t)} = \sum_{\substack{i \in b \\ i \notin \mathcal{C}}} \sum_{k=1}^{|V|} \big[\mathbf{p}^{\mathrm{post},(b,i)}\big]_k \log \frac{\big[\mathbf{p}^{\mathrm{post},(b,i)}\big]_k}{\big[\hat{\mathbf{p}}^{\,(b,i)}\big]_k}. \tag{12}$$

Clean-path cross-entropy (auxiliary).

$$\mathcal{L}_{\mathrm{CE}}^{(b,t)} = \sum_{\substack{i \in b \\ i \notin \mathcal{C}}} \mathrm{CE}\big(\mathbf{x}_0^{(b),i}, \; p_\theta(\hat{\mathbf{x}}_0^{(b),i} \mid c_b)\big). \tag{13}$$

Connectivity penalty (graph validity, incl. block seam).

$$\mathcal{L}_{\mathrm{CN}}^{(b,t)} = -\sum_{i} \sum_{(\hat{v}_0^{(b),i},\, \hat{v}_0^{(b),i+1}) \in E} p_\theta(\hat{v}_0^{(b),i} \mid c_b) \, \log p_\theta(\hat{v}_0^{(b),i+1} \mid c_b). \tag{14}$$

Total objective.

$$\mathcal{L}(\theta) = \mathbb{E}_{(x_0,O,D)\sim\mathcal{P}}\ \mathbb{E}_{t\sim\mathcal{U}[1,T]}\ \mathbb{E}_{\mathbf{x}_t^{(b)}\sim q(\cdot|\mathbf{x}_0^{(b)})} \sum_{b=1}^{B} \Big[\mathcal{L}_{\mathrm{KL}}^{(b,t)} + \lambda\,\mathcal{L}_{\mathrm{CE}}^{(b,t)} + \mu\,\mathcal{L}_{\mathrm{CN}}^{(b,t)}\Big]. \quad (15)$$